Table 3. Comoros: Public Sector Debt Sustainability Framework, Baseline Scenario, 2015-2038 (in percent of GDP, unless otherwise indicated)

	Actual					Estimate	Projections								
	2015	2016	2017	Average 5/	Standard Deviation 5/	2018	2019	2020	2021	2022	2023	2018-23 Average	2028	2038	2024-38 Average
Public sector debt 1/	22.8	27.5	29.3			27.1	27.3	27.5	27.8	28.2	28.8		32.0	38.5	
of which: foreign-currency denominated	22.8	26.4	28.8			27.0	27.3	27.5	27.8	28.2	28.8		32.0	38.5	
Change in public sector debt	3.6	4.7	1.8			-2.1	0.2	0.2	0.2	0.4	0.6		0.6	0.5	
Identified debt-creating flows	-2.6	4.2	-4.0			-0.9	3.7	4.2	4.4	4.7	5.1		6.2	6.9	
Primary deficit	-5.9	5.0	-1.2	-3.2	6.5	1.6	5.1	5.6	5.8	6.1	6.4	5.1	7.6	8.5	7.8
Revenue and grants	31.6	23.4	28.5			28.3	25.5	25.7	26.0	26.3	26.6		27.5	28.0	
of which: grants	15.1	8.9	11.7			10.6	8.6	8.6	8.5	8.5	8.6		8.5	8.4	
Primary (noninterest) expenditure	25.7	28.4	27.3			29.9	30.6	31.3	31.8	32.3	33.0		35.1	36.5	
Automatic debt dynamics	3.3	-0.9	-2.8			-2.6	-1.4	-1.4	-1.4	-1.4	-1.3		-1.4	-1.5	
Contribution from interest rate/growth differential	-0.4	-0.9	-1.0			-1.2	-1.1	-1.2	-1.2	-1.2	-1.2		-1.4	-1.5	
of which: contribution from average real interest rate	-0.2	-0.4	-0.3			-0.4	-0.4	-0.4	-0.4	-0.4	-0.3		-0.4	-0.3	
of which: contribution from real GDP growth	-0.2	-0.5	-0.7			-0.8	-0.7	-0.8	-0.8	-0.9	-0.9		-1.0	-1.2	
Contribution from real exchange rate depreciation	3.7	0.0	-1.7			-1.4	-0.3	-0.2	-0.2	-0.1	-0.1		...	...	
Other identified debt-creating flows	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	6.2	0.5	5.7			-1.2	-3.5	-4.0	-4.2	-4.3	-4.5		-5.5	-6.4	
Other Sustainability Indicators															
PV of public sector debt	...	...	17.6			15.6	15.7	16.1	16.7	17.4	18.2		22.3	28.4	
of which: foreign-currency denominated	...	...	17.1			15.4	15.7	16.1	16.7	17.4	18.2		22.3	28.4	
of which: external	...	...	17.1			15.4	15.7	16.1	16.7	17.4	18.2		22.3	28.4	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	-5.6	5.6	-0.6			3.2	6.0	6.5	6.6	6.8	7.1		8.6	10.4	
PV of public sector debt-to-revenue and grants ratio (in percent)	...	...	61.7			55.2	61.6	62.7	64.2	66.3	68.7		81.0	101.2	
PV of public sector debt-to-revenue ratio (in percent)	...	...	104.6			88.3	92.8	93.9	95.8	98.4	101.3		117.2	144.2	
of which: external 3/	...	...	101.5			87.4	92.8	93.9	95.8	98.4	101.3		117.2	144.2	
Debt service-to-revenue and grants ratio (in percent) 4/	1.1	2.3	2.0			5.3	3.6	3.3	2.9	2.7	2.5		3.8	6.9	
Debt service-to-revenue ratio (in percent) 4/	2.0	3.7	3.4			8.6	5.5	5.0	4.4	4.0	3.7		5.5	9.9	
Primary deficit that stabilizes the debt-to-GDP ratio	-9.5	0.4	-3.0			3.8	4.9	5.4	5.6	5.6	5.9		6.9	7.9	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	1.0	2.2	2.7	2.1	0.8	2.8	2.8	2.9	3.1	3.2	3.3	3.0	3.3	3.3	3.3
Average nominal interest rate on forex debt (in percent)	0.6	0.5	0.2	0.5	0.2	1.1	0.7	0.8	0.9	0.9	1.0	0.9	1.4	1.7	1.5
Average real interest rate on domestic debt (in percent)	...	...	...			...	...	...	...	...	...		...	...	
Real exchange rate depreciation (in percent, + indicates depreciation)	19.4	0.1	-6.8	1.8	8.1	-5.1	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	1.6	2.4	1.2	3.2	1.4	2.4	2.2	2.3	2.3	2.3	2.3	2.3	2.6	2.6	2.6
Growth of real primary spending (deflated by GDP deflator, in percent	7.0	12.8	-1.4	1.9	4.5	12.7	5.0	5.5	4.5	5.0	5.4	6.4	3.7	3.8	4.0
Grant element of new external borrowing (in percent)	...	...	...	...	...	49.8	49.1	42.0	36.1	35.1	35.1	41.2	35.1	35.1	...

Sources: Country authorities; and staff estimates and projections.

1/ [Indicate coverage of public sector, e.g., general government or nonfinancial public sector. Also whether net or gross debt is used.]

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

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Table 4. Comoros: Sensitivity Analysis for Key Indicators of Public Debt 2018-2038
(baseline scenario)

	Projections							
	2018	2019	2020	2021	2022	2023	2028	2038
PV of Debt-to-GDP Ratio								
Baseline	16	16	16	17	17	18	22	28
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	16	11	5	1	-4	-9	-38	-94
A2. Primary balance is unchanged from 2018	16	14	11	9	8	6	-7	-31
A3. Permanently lower GDP growth 1/	16	16	16	17	18	19	24	33
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2019-20	16	16	17	18	19	20	26	35
B2. Primary balance is at historical average minus one standard deviations in 2019-202	16	15	13	14	15	16	20	27
B3. Combination of B1-B2 using one half standard deviation shocks	16	13	10	11	12	13	19	29
B4. One-time 30 percent real depreciation in 2019	16	22	21	21	22	22	25	29
B5. 10 percent of GDP increase in other debt-creating flows in 2019	16	22	22	23	23	24	28	32
PV of Debt-to-Revenue Ratio 2/								
Baseline	55	62	63	64	66	69	81	101
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	55	41	21	2	-16	-35	-135	-314
A2. Primary balance is unchanged from 2018	55	53	44	37	29	21	-27	-109
A3. Permanently lower GDP growth 1/	55	62	63	65	67	70	86	117
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2019-20	55	63	66	69	72	75	93	123
B2. Primary balance is at historical average minus one standard deviations in 2019-202	55	57	52	54	57	59	73	96
B3. Combination of B1-B2 using one half standard deviation shocks	55	49	37	41	44	48	68	103
B4. One-time 30 percent real depreciation in 2019	55	85	83	83	83	84	90	102
B5. 10 percent of GDP increase in other debt-creating flows in 2019	55	87	87	88	89	91	100	113
Debt Service-to-Revenue Ratio 2/								
Baseline	5	4	3	3	3	3	4	7
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	5	4	3	2	1	0	-1	-15
A2. Primary balance is unchanged from 2018	5	4	3	2	2	2	1	-4
A3. Permanently lower GDP growth 1/	5	4	3	3	3	3	4	8
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2019-20	5	4	3	3	3	3	4	8
B2. Primary balance is at historical average minus one standard deviations in 2019-202	5	4	3	3	2	2	4	6
B3. Combination of B1-B2 using one half standard deviation shocks	5	4	3	2	2	2	4	6
B4. One-time 30 percent real depreciation in 2019	5	4	5	4	4	4	6	11
B5. 10 percent of GDP increase in other debt-creating flows in 2019	5	4	4	4	3	3	4	8

Sources: Country authorities; and staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.
2/ Revenues are defined inclusive of grants.

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